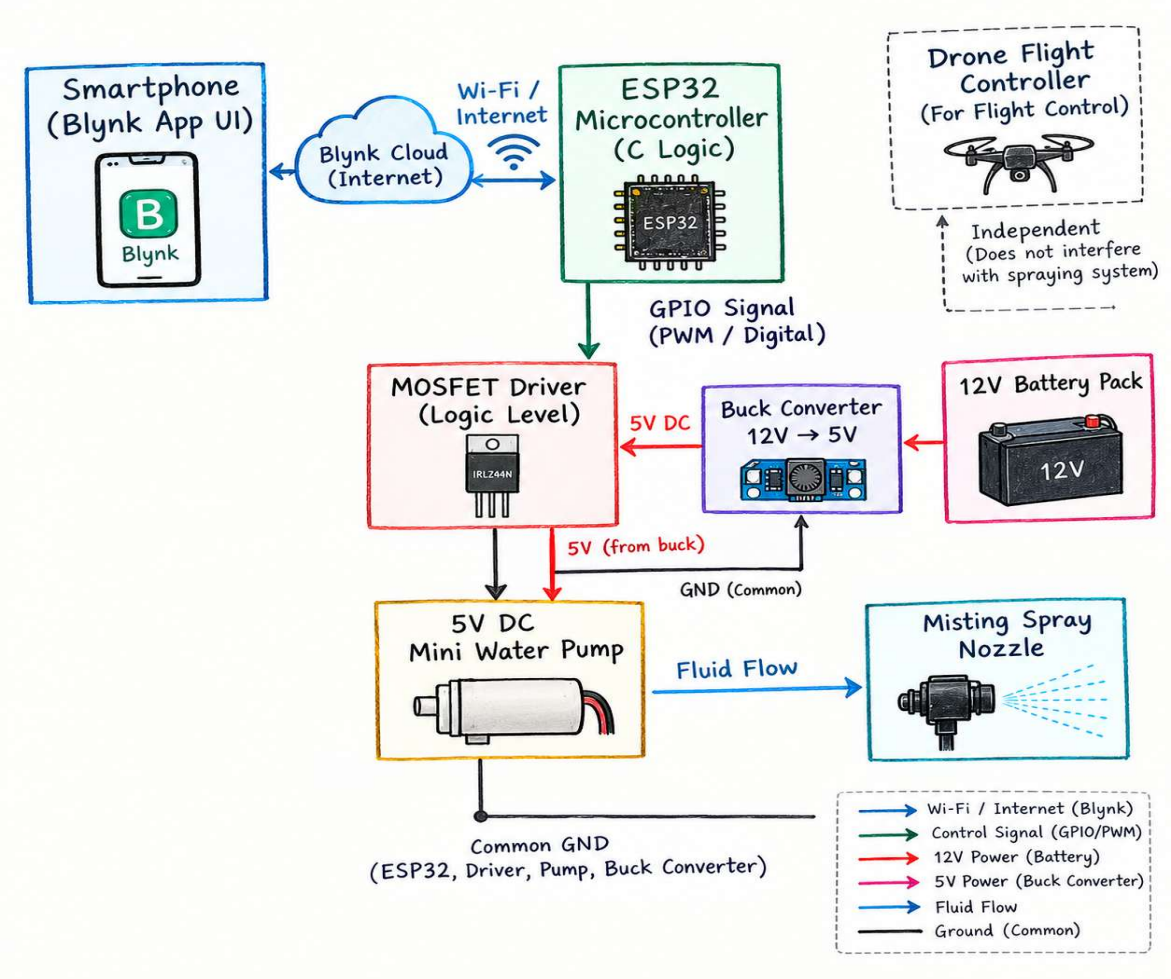
Methodology Our project develops a modular, IoT-driven spraying payload that retrofits onto a custom ₹8,000 F450 quadcopter. We fundamentally decouple the spraying mechanism from expensive, proprietary flight controllers to maximize accessibility. The payload's processing core is an ESP32 microcontroller programmed in C, which establishes a localized Wi-Fi hotspot to communicate directly with the operator's smartphone via the Blynk application.

When the user adjusts the interface slider, the app transmits data packets to the ESP32. The embedded control flow logic translates these inputs into Pulse Width Modulation (PWM) signals. Because the 3.3V microcontroller cannot safely drive high-current hardware directly, the PWM signal routes through an L298N motor driver. This driver regulates power from an isolated LiPo battery to a DC mini-water pump. By dynamically altering the pump's motor speed, the system achieves Variable Rate Application (VRA)—allowing precise software control over the fertilizer fluid pushed through silicone tubing to the misting nozzle.

Proposed Block Diagram



Proposed Outcome

The project delivers a fully functional, ultra-low-cost smart spraying prototype. By integrating fundamental electronic components and IoT architecture, we successfully shift payload control from complex RC transmitters to an intuitive smartphone interface.

The primary outcome is the democratization of precision agriculture. At a total system cost of just ₹8,000, small-scale farmers gain access to Variable Rate Application technology previously restricted to commercial drones costing lakhs of rupees. This innovation ensures targeted fertilizer distribution based on crop density, drastically reducing chemical waste and soil degradation. Furthermore, by enabling remote operation via Wi-Fi, the system physically distances workers from toxic chemical drift, eliminating severe respiratory and dermatological health risks. Ultimately, the prototype provides a highly modular, scalable framework ready for future automated sensor integration without requiring a hardware overhaul.